

Question 25 (2 points)

25 Solve the equation below algebraically.

$$6 - \frac{2}{3}(x + 5) = 4x$$

Answer	$\frac{4}{7}$ or 0.571428 or equivalent		
Expectation Level	Level 1		
Content Tag	One-variable linear equations		
Points	0	1	2
Description	Response is incorrect due to two or more errors.	Response is correct without algebraic work shown. OR Response is incorrect due to one error. Examples of errors include: - One mistake with expanding or simplifying expressions - One operation is incorrect (i.e, scholar multiplies by $\frac{14}{3}$) - Rounding the value instead of representing it as a repeating decimal	Response is correct with algebraic work shown.

Question 26 (2 points)

26 $F = 4x^2 - 3x + 1$ and $G = -2x^2 + 5$. Find the trinomial that represents H , if $H = 3G - F$.

Answer	$-10x^2 + 3x + 14$ or equivalent		
Expectation Level	Level 2		
Content Tag	Equivalent expressions		
Points	0	1	2
Description	Response is incorrect due to two or more errors.	Response is correct with no work shown. OR Response is incorrect due to one error. Examples of errors include: - One operation is incorrect (i.e, scholar distributes the “-” incorrectly) - One computational error - Trinomial is not fully simplified	Response is correct with work shown.

Question 27 (2 points)

27 Algebraically determine the zeros of the function $f(x) = x^2 - 4x + 3$.

Answer	1 and 3		
Expectation Level	Level 1		
Content Tag	Quadratic functions		
Points	0	1	2
Description	Response is incorrect due to two or more errors.	Response is correct with no work shown. OR Response is incorrect due to one error. Examples of errors include: - One computational error - One factoring error - Zeros stated as coordinates - Non-algebraic method	Response is correct with algebraic work shown. Acceptable algebraic strategies include: - Factoring - Complete the square - Quadratic formula Algebraic work should show: - Setting $f(x) = 0$ - All new equivalent equations created at each step

Question 28 (2 points)

In February 2021, the state of Texas saw one of the worst snowstorms in its history. Snowfall totals recorded in Dallas, TX are shown in the table below.

Dallas, TX	
Time	Snow (inches)
1 a.m.	1
3 a.m.	5
6 a.m.	11
12 noon	33
3 p.m.	36

Which of the following intervals had the greatest rate of snowfall, in inches per hour? 1 a.m. to 12 noon or 6 a.m. to 3 p.m. Justify your answer.

Answer	1 a.m. to 12 noon		
Expectation Level	Level 1		
Content Tag	Features of linear equations		
Points	0	1	2
Description	Response is correct without justification. OR Response is incorrect due to two or more errors.	Response is incorrect due to one error. Examples of errors include: - One computational error - One mistake interpreting the data or their own calculations	Response is correct with sufficient justification. In order for justification to be considered complete, scholars must: - Clearly state which interval has the greater rate of snowfall. - Calculate or estimate the rate of snowfall for both intervals.

Question 29 (2 points)

- 29** The gravitational force between two objects can be calculated using the formula $F_g = \frac{GM_1M_2}{r^2}$. In the formula, G is called the gravitational constant. The variables M_1 and M_2 represent the masses of the two objects between which the force is being measured. Lastly, r is the distance between the objects. Solve for the positive value of r , in terms of the other variables.

Answer	$r = \sqrt{\frac{GM_1M_2}{F_g}}$		
Expectation Level	Level 2		
Content Tag	Equivalent expressions		
Points	0	1	2
Description	Response is incorrect due to two or more errors.	Response is incorrect due to one error. Examples of errors include: - One operation is incorrect - Solving for a value other than r Ex. $r^2 = \frac{GM_1M_2}{F_g}$ - Giving both the positive and negative value of r Ex. $r = \pm \sqrt{\frac{GM_1M_2}{F_g}}$	Response is correct with sufficient work shown. Note: Variables must be used as they are written in the problem. (i.e, r cannot be written in uppercase.)

Question 30 (2 points)			
30 Four Seasons Landscaping is creating a rectangular flower bed at a customer's home. The width of the flower bed will be half of the length. The total area of the flower bed is 38 square feet. Write and solve an equation to determine the width of the flower bed to the <i>nearest tenth</i> of a foot.			
Answer	$2w^2 = 38$ or equivalent 4.4 feet		
Expectation Level	Level 2		
Content Tag	Quadratic functions		
Points	0	1	2
Description	Response is incorrect due to two or more errors.	Equation and/or response is correct with no work shown. OR Response is incorrect due to one error. Examples of errors include: • One computational error • Missing equation • Correctly solving for the length of the flower bed (8.7 feet)	Equation and response are correct with sufficient work shown.

Question 31 (4 points)

A couple planning a backyard wedding is researching different tent rental companies for their event. They contact two different companies. Buffalo Party Rental charges a \$50 set-up fee, and \$5 per square foot of tent space. Tent Genie charges a \$25 set-up fee, and \$6 per square foot of tent space.

Write an equation for Buffalo Party Rental that could be used to determine the total cost B , when x square feet of tent space is ordered. Write a second equation for Tent Genie that could be used to determine the total cost T , when x square feet of tent space is ordered.

Determine algebraically and state the *minimum* square feet, rounded to the *nearest square foot*, of tent space that must be ordered for Buffalo Party Rental to be the cheaper option.

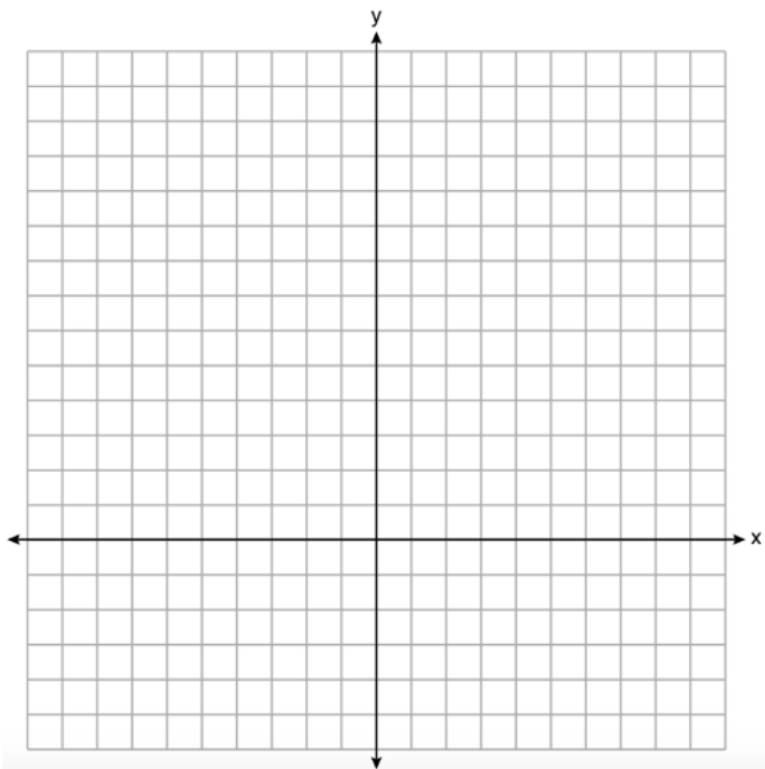
Answer	$B = 5x + 50$, $T = 6x + 25$		
	26		
Expectation Level	Level 2		
Content Tag	System of Equations		
Points	0	1	2
System of Equations (2 points)	Response is incorrect.	One equation is correct.	Both equations are correct.
Answer (2 points)	Response is incorrect due to two or more errors.	Response is correct with no work shown. OR Response is incorrect due to one error. Examples of errors include: - Using a non-algebraic strategy. - Making one computational error. - One error interpreting the solution	Response is correct or matches equations from Part A with algebraic work.

Question 32 (4 points)

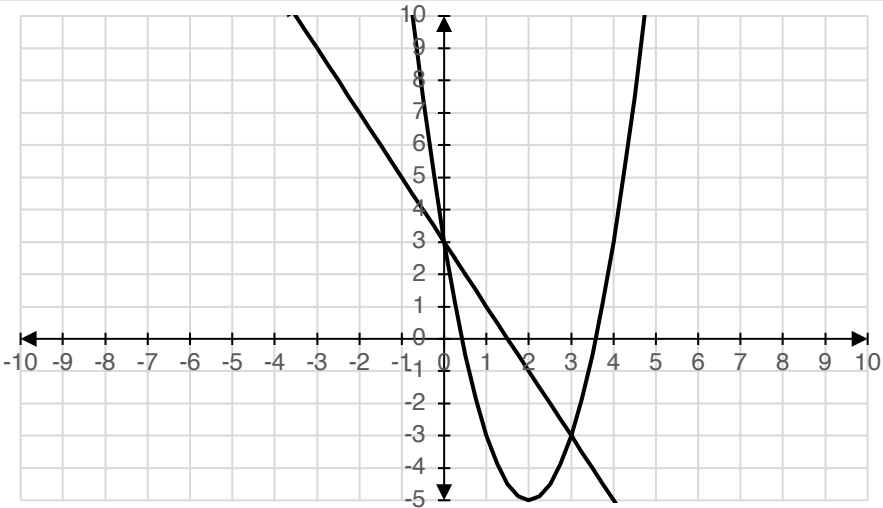
32 Graph the two functions on the coordinate plane below.

$$f(x) = 2x^2 - 8x + 3$$

$$g(x) = -2x + 3$$



Find all values of x for which $f(x) = g(x)$.

Answer			
	0 and 3		
Expectation Level	Level 2		
Content Tag	System of equations		
Points	0	1	2
Graph (2 points)	Graph is incorrect due to two or more errors.	Graph is incorrect due to one error. Examples of errors include: - One graphing error - Graphing only one function correctly	Graph is correct. Graph must include the following to be considered correct: - The vertex (2, -5) and two other distinct points are plotted accurately for the parabola - The two functions intersect at (0,3) and (3, -3) - The line and parabola both have arrows or extend to the edge of the graph.
Solutions (2 points)	Response is incorrect due to two or more errors.	Response is correct with no work shown. OR Response is incorrect due to one error. Examples of errors include: - One computational error - Stating the solutions as (0,3) and (3, -3) - Stating only one solution	Response is correct or matches graph in Part A with work shown.

Question 33 (4 points)

- 33** The table below shows the number of hours ten different scholars at Marion P. Thomas Middle School spent studying and their scores on a recent history test.

Hours Spent Studying (x)	0	1	2	4	4	4	6	6	7	8
Test Scores (y)	35	40	46	65	67	70	82	88	82	95

Write the linear regression for this data set. Round all values to the *nearest hundredth*.

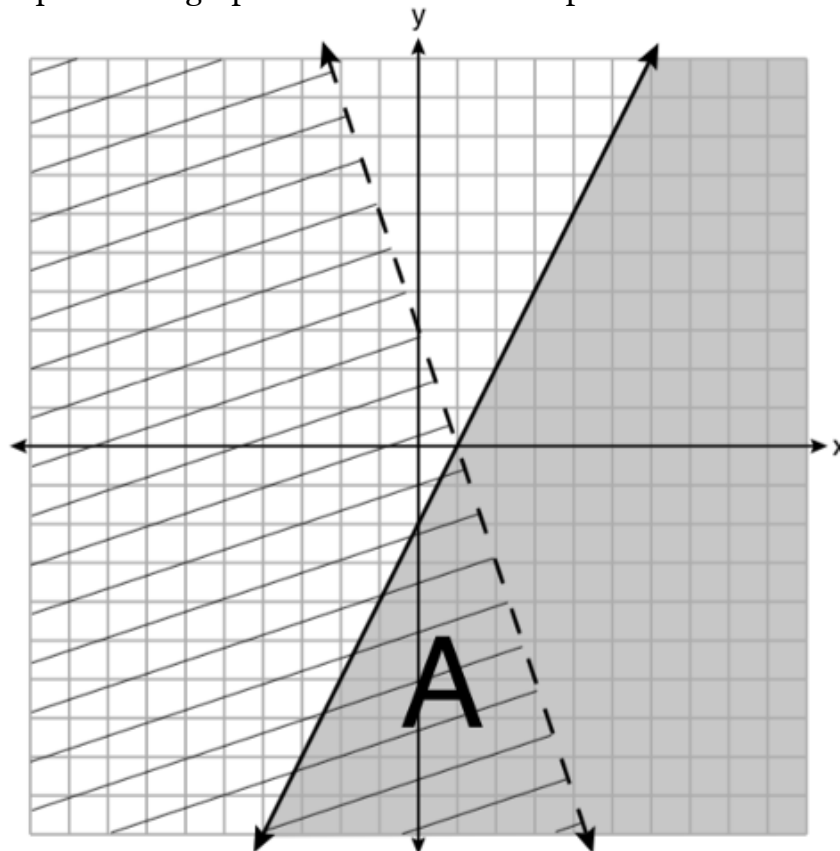
State the correlation coefficient rounded to the *nearest hundredth*.

Explain what the correlation coefficient suggests in the context of the problem.

Answer	$y = 7.79x + 34.27$		
	0.98		
	Explanations may vary. For example, “The correlation coefficient suggests that there is a strong, positive association between time spent studying and test scores.”		
Expectation Level	Level 1		
Content Tag	Linear regression		
Points	0	1	2
Regression Equation (2 points)	Response is incorrect due to two or more errors.	Equation is incorrect due to one error. Examples of errors include: - One rounding error - Writing an expression instead of an equation	Equation is correct.
Correlation coefficient (1 point)	Response is incorrect.	Response is correct.	
Explanation (1 point)	Explanation is incorrect. Note: If a scholar does not give any correlation coefficient, they cannot receive any credit for the explanation because it is impossible to tell if they are interpreting a value correctly when that value is missing.	Explanation is complete and accurately interprets scholar’s stated correlation coefficient. In order for the explanation to be considered complete, scholars must: - Refer to both the hours spent studying and the test scores. - State whether the correlation is positive or negative. Note that a scholar can state this indirectly by stating how one value changes as the other changes. Ex. “Students who spend more hours studying have higher test scores.”	

Question 34 (4 points)

34 A system of inequalities is graphed on the coordinate plane below.



Write the system of inequalities that is represented by the graph above.

What does region A represent?

What does the gray region represent?

Answer	$y < -3x + 3$ and $y \leq 2x - 2$ or equivalent		
	Region A represents the solution to the system of inequalities.		
	The entire gray region represents the solution to $y \leq 2x - 2$.		
Expectation Level	Level 1		
Content Tag	System of inequalities		
Points	0	1	2
System of Inequalities (2 points)	Response is incorrect.	One inequality is correct.	Both inequalities are correct.
Region A (1 point)	Response is incorrect.	Response is correct.	
Gray Region (1 point)	Response is incorrect.	Response is correct.	

Question 35 (6 points)

- 35** The CoinStar machine at the local supermarket is broken. Instead of providing the exact number of each type of coin that is deposited, the machine can only display the total number of coins and total value. The machine shows 90 coins, with a value of \$17.55.

If Mr. Rogers only placed dimes and quarters in the machine, write a system of equations in two variables, or an equation in one variable that could be used to model this situation.

Using your equation or system of equations, algebraically determine the number of quarters Mr. Rogers deposited in the machine.

Mr. Rogers' neighbor replaced each one of his dimes with a quarter. If he uses all of his coins, determine if Mr. Rogers would have enough money to buy a cardigan priced at \$20.98 if he must also pay an 8% sales tax. Justify your answer.

Answer	$d + q = 90$, $0.10d + 0.25q = 17.55$ or equivalent system OR $0.25q + 0.10(90 - q) = 17.55$ or equivalent		
	57		
	Mr. Rogers will not have enough money.		
Expectation Level	Level 2		
Content Tag	Systems of equations		
Points	0	1	2
Equation / System of equations (2 points)	Response is incorrect.	Response is incorrect due to one error. Examples of errors include: Writing only one correct equation with two variables	Response is correct.
Solution (2 points)	Response is incorrect due to two or more errors.	Response is correct with no algebraic work shown. OR Response is incorrect due to one error. Examples of errors include: - One computational error - One operation is incorrect	Response is correct or matches equation(s) from part A with algebraic work shown.
Cardigan (2 points)	Response is incorrect due to two or more errors.	Response is incorrect due to one error. Examples of errors include: - Incomplete justification - One computational error - One rounding error	Response is correct with complete justification. In order for justification to be considered complete, scholars must: - Calculate Mr. Roger's new total (\$22.50) - Calculate the tax for cardigan or cost of cardigan (\$1.68 or \$22.66) - Compare those values